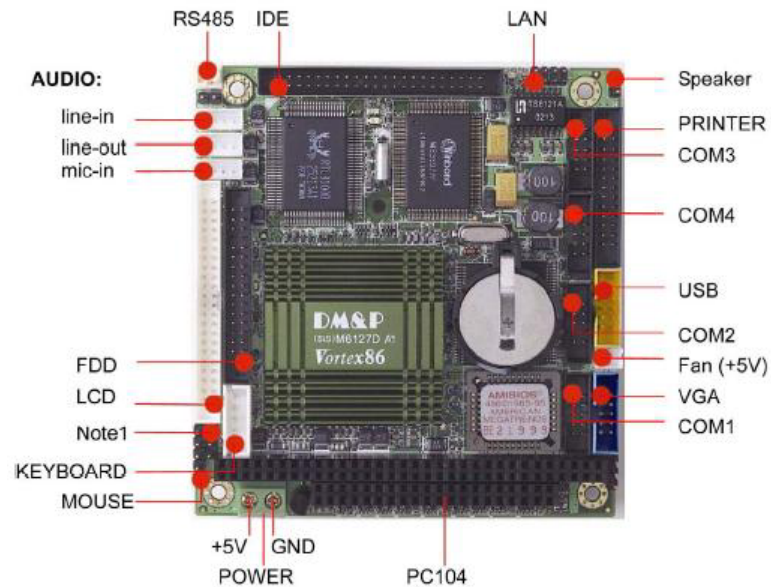


ภาคผนวก ข.

รายละเอียดอุปกรณ์

ข.1 รายละเอียดของบอร์ดประมวลผล Vortex86-6071

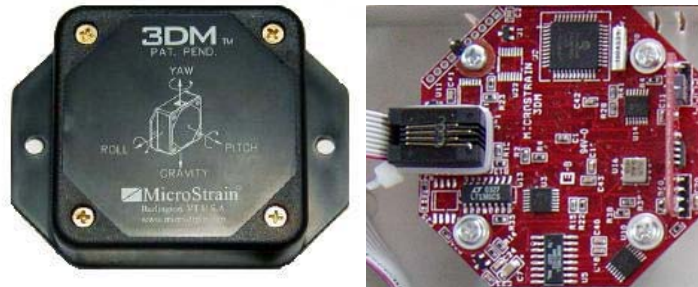


รูปที่ ข.1 บอร์ดประมวลผล Vortex86-6071

Processor Chipset	DM&P(SiS)Vortex86 . System-on-Chip CPU-200MHz Real Time Clock with Lithium Battery Backup
Bus Interface	ISA and PC/104 Standard Compliant
Memory	Onboard 128MB SDRAM (optional 128MB)
BIOS	AMI BIOS
Multi I/O	Enhanced IDE interface RS232 port x3 RS232/485 port x1 Parallel port x1 FDD interface x1 USB port x1 Line-in port x1 Line-out port x1 Mic-in port x1
Video Display	AGP Rev.2.0 Compliant

	Shared system memory area up to 16MB.
	Resolution up to 1,920Cx 1,440 true colors
	CRT/LCD display
LAN	Realtek 8100B single chip
	Full-duplex transfer mode, doubles effective bandwidth 16KB RAM buffer
	Throughput 10/100Mbps
Connectors	2.0mm 44-pin box header for IDE
	2.0mm 10-pin box header for RS-232 x3
	2 pin header for RS-485
	2.0mm 26-pin box header for Printer
	2.0mm 34-pin box header for FDD
	2.0mm 10-pin box header for USB
	2.0mm 10-pin box header for VGA
	2.0mm 8-pin header for Ethernet
	2.0mm 44-pin box header for LCD connector
	2.0mm 4-pin header for Line-in
	2.0mm 4-pin header for Line-out
	2.0mm 4-pin header for Mic-in
	5-pin box header for AT-Keyboard connector
	5-pin header for PS/2 Mouse
Watchdog Timer	Software Watchdog Timer
	Three 8254 Compatible Programmable 16-bit Counters.
Power Requirement	Single Voltage +5V @1.2 A
Board Weight	100g
Board Size	90mm X 96mm
Operating Temperature	-20°C ~ +70°C

ข.2 รายละเอียดของอุปกรณ์วัดความเร่ง 3 แกน (3DM)



รูปที่ ข.2 อุปกรณ์วัดความเร่ง 3 แกน (3DM)

Range	Yaw: ± 180 degrees
	Pitch: ± 180 degrees
	Roll: ± 70 degrees
A/D Resolution	12 bits
Digital Filter	Infinite Impulse Response (IIR) User programmable
	weighted moving average
Angle Resolution (no digital filtering)	Pitch: 0.30 degrees (typical)
	Roll: 0.25 degrees (typical)
	Yaw: 0.50 degree (typical)
Angle Resolution (most aggressive digital filtering)	Pitch: < 0.1 degrees
	Roll: < 0.1 degrees
	Yaw: < 0.1 degrees
Resolution specs. taken during static motions.	
Accuracy	Pitch: ± 0.93 degree typical (yaw from 0 - 360 degrees & roll=0 degrees)
	Roll: ± 0.33 degree typical (yaw from 0 - 360 degrees & pitch =0 degrees)
	Yaw: ± 1.0 degrees typical (pitch & roll=0 degrees)

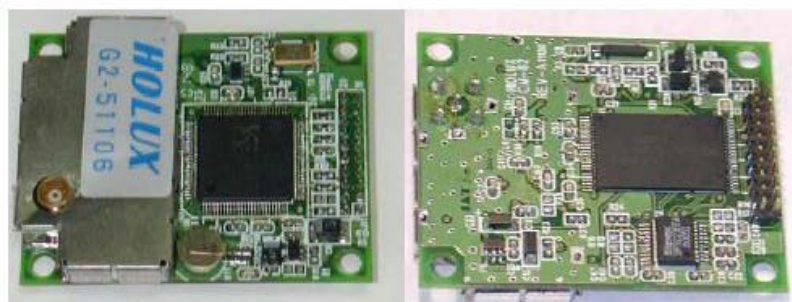
Accuracy is defined as the square root of the sum of the errors squared (non repeatability, temperature coefficients & non-linearity).

Angle measurement non-linearity (pitch & roll)	$\pm 0.23\%$ F.S.
Angle measurement repeatability	Pitch: 0.07 degrees (typical) Roll: 0.07 degrees (typical) Yaw: 0.26 degrees (typical)
Update rate (angle mode)	45 Hz/ 3 channels (maximum) 30 Hz/ 3 channels (typical)

The update rate is specified with a maximum and typical value since it depends on how many points the A/D converter averages.

Update rate (raw mode)	70 Hz/ 6 channels
Output modes	Raw: ax, ay, az accelerometer Raw: mx, my, mz magnetic field Units: pitch, roll, & yaw in degrees
Output format	RS-232 serial or RS-485 multi-drop network
Transmission Rate	9600 bits/sec
Supply voltage	5.3 volts DC min
Supply current	50 milliamps/node @ standard speed
Connectors	Sensor: RJ11 type power: min. coaxial jack
Operating Temperature	- 25 degrees C to 70 degrees C
Temperature Drift (%/ deg. C) (mean, std.dev.)	Pitch: 0.009+/-0.008 Roll: 0.033+/-0.025 Yaw: 0.019+/-0.019
Module size	1.7" wide, 2.5" long, 0.7 " thick
3DM enclosure	2.05" wide x 3.815" long x 0.9" thick

ข.3 รายละเอียดของ GPS Receiver (Holux รุ่น GM-82)



รูปที่ ข.3 GPS Receiver (Holux รุ่น GM-82)

1. SiRF Star II chipset with embedded ARM7TDMI CPU available for customized applications in firmware
2. 12 parallel satellite-tracking channels for fast acquisition and reacquisition
3. High speed signal acquisition using 1920 time/frequency search channels
4. Use Frequency 1575.42 MHz
5. Built-in WAAS/EGNOS Demodulator
6. Support U.S. Coast Guard DGPS beacon signal
7. Low power consumption with Advanced Trickle-Power and Push-To-Fix mode
8. Optional Rechargeable battery for memory and RTC backup and for fast Time to First Fix(TTFF)
9. Support NMEA0183 v2.2 data protocol and SiRF binary code
10. Enhanced algorithms -SnapLock and SnapStart provide superior navigation performance in urban, canyon and foliage environments
11. For Car Navigation , Marine Navigation ,Fleet Management ,AVL and Location-Based Services ,Auto Pilot ,Personal Navigation or touring devices, Tracking devices/systems and Mapping devices application.
12. Input voltage: 5.0Vdc +/-10% or 3.3Vdc +/- 10%
13. Backup power:(optional) ◦ 3V Rechargeable Lithium cell battery, up to 1000 hours discharge.
14. MCX antenna connector: Active or patch.
15. PCB Size: 40(W) x 50(D) x 12(H) (mm)
16. Weight: 18 g

Performance

1. Update rate: 1 second.
2. Acquisition time:
 - Reacquisition 0.1 sec, averaged
 - Hot start 8 sec. averaged
 - Warm start 38 sec. averaged
 - Cold start 45 sec. averaged
3. Position accuracy:
 - Non DGPS (Differential GPS)
 - Position 5~25 meter CEP
 - Velocity 0.1 meters/second.
 - Time 1 microsecond synchronized GPS time
 - DGPS (Differential GPS)
 - Position 1 to 5 meters, typical
 - Velocity 0.05 meters/second, typical
 - EGNOS/WAAS
 - Position < 2.2 meters, horizontal 95% of time
< 5 meters, vertical 95% of time